19L820-Project Work 2

BATCH NO:19

Deep Learning based Channel Estimation For OFDM Wireless Communication

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PROBLEM STATEMENT

Channel State Information (CSI) is critical for the performance of **OFDM-based wireless communication systems**. Traditional estimation methods, **Least Squares (LS)** and **Linear Minimum Mean Squared Error (LMMSE)**, face significant challenges:

- **Sensitivity to Noise:** Traditional methods often suffer from severe performance degradation in **low SNR** environments.
- **Statistical Dependency:** LMMSE requires **prior knowledge of channel statistics**, which are often unavailable in real-world scenarios.
- **Limitations of Existing DL Models: CNNs** offer improvements, they often fail to capture **long-range time-frequency correlations**

By these limitations developing a software-based framework that cascades an **Initial Denoising Network (IDN)** with a **Transformer-based model**. This approach aims to suppress pilot noise and exploit **global correlations** across the OFDM grid to reconstruct high-fidelity CSI without requiring prior channel statistics.

LITERATURE SURVEY 1

Deep-Learning Based Channel Estimation for OFDM Wireless Communications

Authors: Guoda Tian, Xuesong Cai, Tian Zhou, Weinan Wang, Fredrik Tufvesson
Journal: IEEE International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)
Year: 2022

Work Done:

- Proposed a deep learning–based channel estimation pipeline for OFDM systems to estimate full CSI from limited pilot symbols **without increasing pilot power** or density.

- Introduced an **Initial Denoise Network (IDN)** and a **Resolution Enhancement Network** (REN) using fully connected layers to improve pilot SNR and recover high-resolution CSI.

- Achieved **15 dB lower NMSE than LMMSE at 0 dB SNR** and **10 dB lower NMSE at 20 dB SNR**. Trained using **210,000 samples** and tested on **21,000 samples (1,000 per SNR point),** with a compact model suitable for embedded systems.

Limitations:

- Evaluated only for **SISO OFDM** systems.

- Requires **supervised training** with large labeled datasets.

LITERATURE SURVEY 2

Deep Learning-Based Channel Estimation with Transformer

Authors: J. Guo, et al.
Journal: IEEE Access (2021)
Year: 2021

Work Done:

- **Emphasized the use of Self-Attention** to effectively capture the complex, non-local time-frequency correlations inherent in wireless channels.

- Demonstrated that the Transformer model outperforms CNN-based models by achieving around **18 dB lower NMSE** than CNN models at a **pilot SNR of 30 dB**

- Confirmed that the Transformer architecture is uniquely suited for processing sequential and global grid data by integrating **Positional Encoding** to preserve the time-frequency structure.

Limitations:

- The implementation is **computationally expensive** due to the high complexity of the self-attention mechanism, potentially increasing training time compared to simpler CNN/MLP models.

- Performance is highly dependent on the **quality of the positional encoding** scheme used.

LITERATURE SURVEY 3

Denoising Generalization Performance of Channel Estimation in Multipath Time-Varying OFDM Systems

Author: Yinying Li, Xin Bian, Mingqi Li

Journal: Sensors (MDPI)

Year: 2023

Work Done:

- Designed a deep network with Denoising CNN (DnCNN) to **suppress AWGN** from pilot-based CSI matrices.
- Modeled the channel frequency response as a 2D grid and applied **residual learning** to refine the initial LS/LMMSE estimate.
- Achieved stable NMSE performance over an SNR range of 0–35 dB, and maintained performance when tested across various channel models; at 30 dB SNR, average pooling achieved 29.49 dB RMSE compared to 28.36 dB with max pooling.

Limitation:

- Depends on the accuracy of the initial LS/LMMSE estimate; **large errors propagate** into the denoiser.
- Requires channel/noise statistics implicitly learned from large **labeled datasets**.
- **Increased computational** and **memory cost** due to deep CNN and residual blocks.

LITERATURE SURVEY 4

ChannelNet: A Deep Neural Network for Channel Estimation in OFDM Systems

Authors: Ye Yuan, Guanding Yu, Hailin Zhang, Yonghui Li
Journal: IEEE Wireless Communications Letters
Year: 2019

Work Done:

- Proposed **ChannelNet**, a CNN-based deep learning framework for OFDM channel estimation.
- Designed a two-stage architecture consisting of a **super-resolution CNN** and an **image restoration CNN** to reconstruct high-quality CSI from sparse pilot signals.
- Treated channel estimation as an image recovery problem by representing CSI as a 2D time–frequency grid.
- Achieved **NMSE performance** comparable **to full MMSE** and **lower NMSE** than **LS and ALMMSE** under LTE simulation settings.

Limitations:

- CNN architecture mainly captures **local spatial features** and struggles to learn long-range time–frequency dependencies.
- Requires **high computational resources** as **20-layer** CNN increases training complexity.
- Performance **depends heavily** on pilot distribution and training **dataset** diversity.

LITERATURE SURVEY 5

A DFT-based Low Complexity LMMSE Channel Estimation Technique for OFDM Systems

Authors: Jyoti Prasanna Patra, Bibhuti Bhusan Pradhan, Poonam Singh

Journal: Journal of Telecommunications and Information Technology (JTIT)

Year: 2022

Work Done

- **Reduced the computational complexity** of conventional LMMSE from **$O(N^3)$** to **$O(N \log N)$** by exploiting DFT properties and structured correlation.
- Achieved near-optimal NMSE/BER performance compared to standard LMMSE and significantly better performance than LS.
- Demonstrated that LMMSE remains a strong and commonly used baseline in OFDM channel estimation.

Limitation

- Still **requires channel statistics / covariance information**, which may not always be accurately available in practical systems.**Performance** depends on the **assumed** channel correlation model.
- Although **complexity** is reduced, it is still **$O(N \log N)$,** which is higher than LS and may be **heavy** for **very low-power devices**..

OBJECTIVE

- To develop a **software-based OFDM** channel estimation framework using deep learning techniques
- To **analyze the limitations** of traditional channel estimation methods under **noisy** and **low-SNR** conditions
- To design and implement a **pilot denoising network** to improve the quality of received channel information
- To apply a **Transformer-based model** for learning global time–frequency channel correlations
- To achieve **improved channel estimation** accuracy compared to conventional and existing **deep learning** approaches
- To evaluate system performance using error metrics such as **Mean Squared Error** (MSE) across different SNR levels

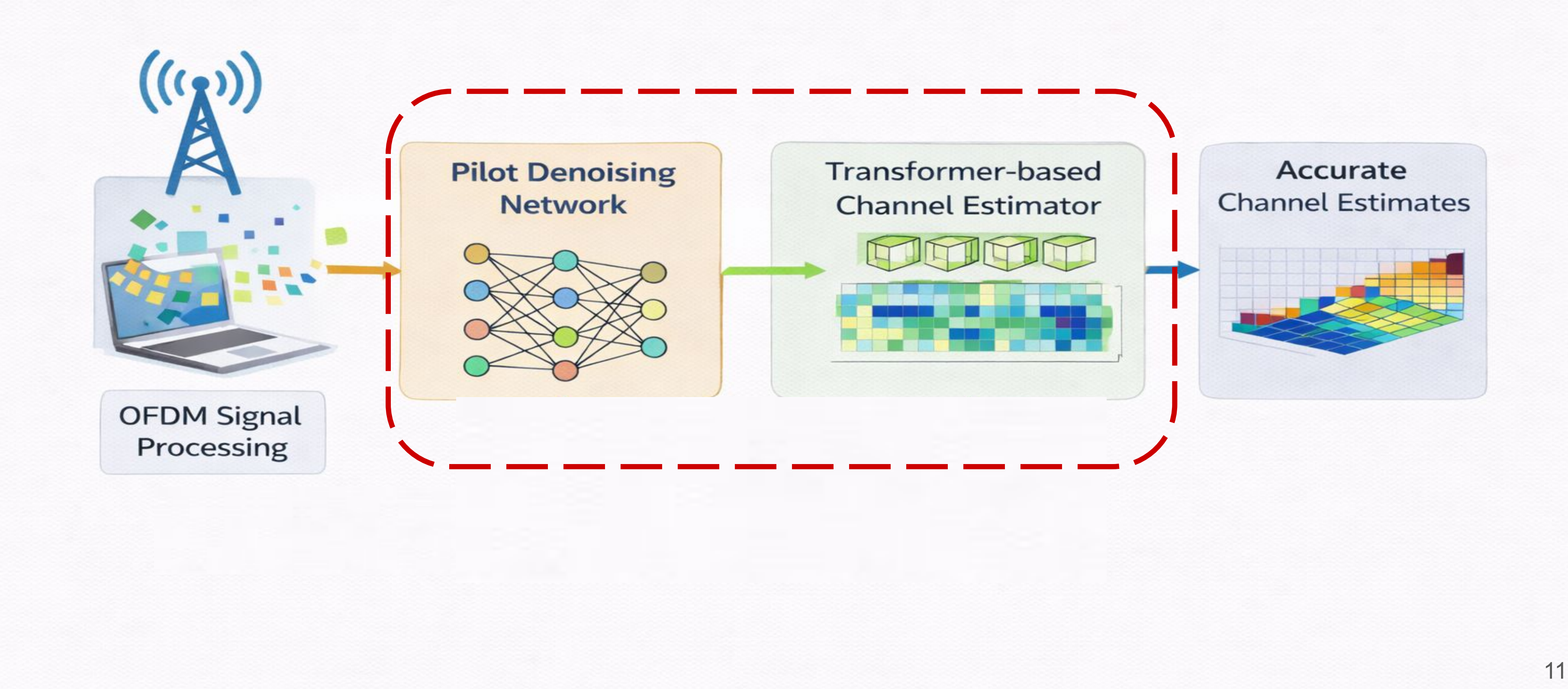
SCOPE OF THE PROJECT

- **Software-Defined Deep Learning Framework:** Engineering a comprehensive software-based estimation pipeline specifically designed for high-fidelity CSI recovery in OFDM wireless systems.
- **Realistic Channel Environment Simulation:** Modeling complex wireless signal conditions utilizing Rayleigh Fading and Additive White Gaussian Noise (AWGN) to analyze estimation robustness.
- **Transformer Architecture:** Implementing a Transformer-based model to exploit multi-head self-attention mechanisms for learning global time-frequency correlations across the resource grid.
- **Quantitative Performance Evaluation:** Assessing system accuracy through standardized MSE and NMSE metrics across a dynamic range of Signal-to-Noise Ratio (SNR) levels.
- **Standardized Benchmarking:** Conducting a comparative analysis to validate the proposed model against conventional Least Squares (LS) and Linear Minimum Mean Squared Error (LMMSE) techniques.
- **Simulation-Based Research Boundaries:** Focus is restricted to high-performance simulation-based modeling using Python, PyTorch, and Google Colab to validate architectural efficiency rather than physical hardware implementation.

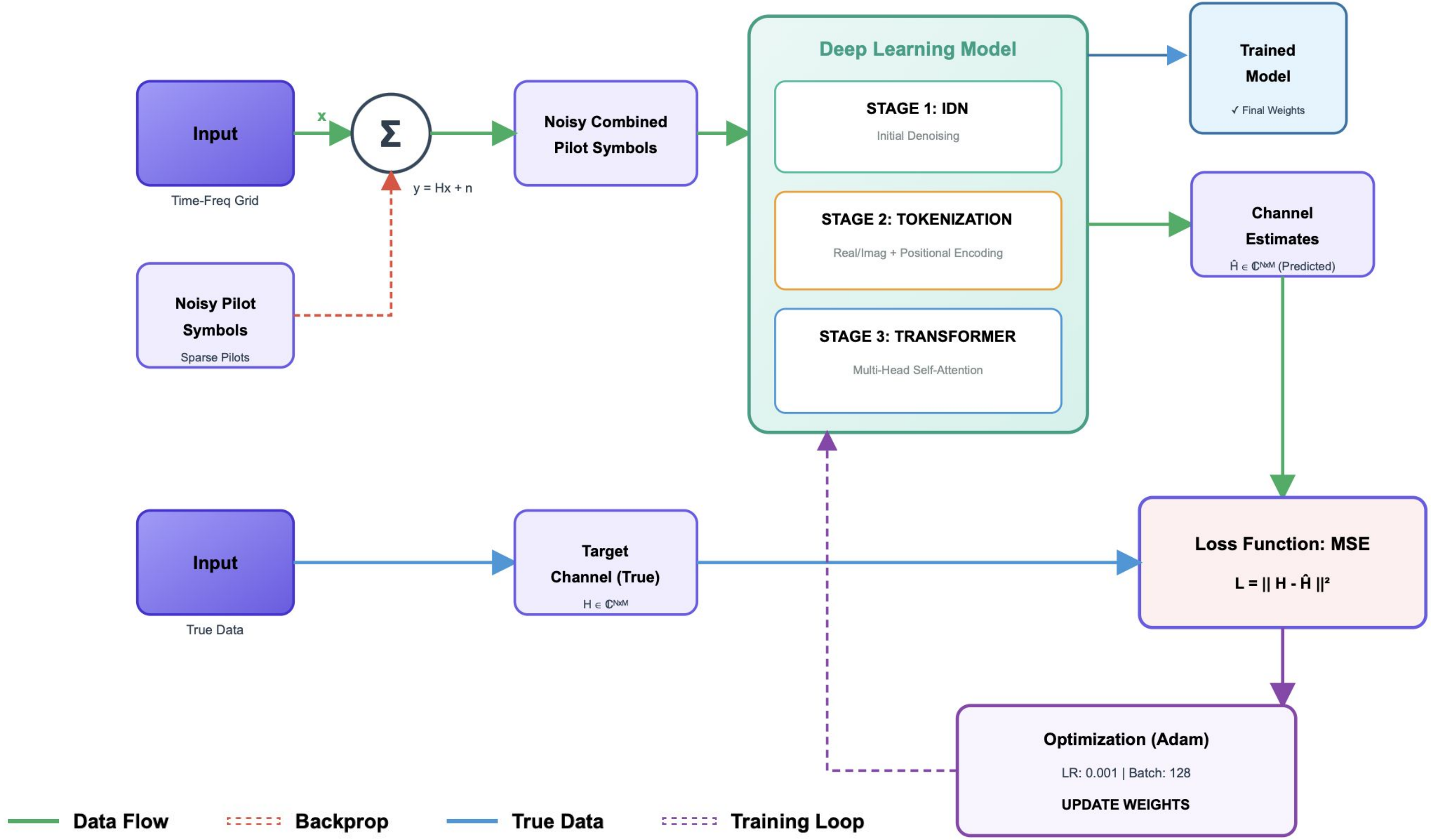
PROPOSED METHODOLOGY

- A software-based deep learning framework is developed for accurate OFDM channel estimation using pilot symbols only.
- The received pilot **Channel State Information (CSI)**, corrupted by noise, is first processed using an **Initial Denoising Network (IDN)** implemented as a fully connected neural network to suppress noise and enhance pilot quality.
- The denoised pilot CSI is then converted into a sequence of tokens by separating real and imaginary components and appending time–frequency positional information.
- A **Transformer-based channel estimation model** is employed to capture long-range correlations across subcarriers and OFDM symbols using self-attention mechanisms.
- The Transformer reconstructs the complete CSI of the OFDM time-frequency grid from the limited pilot information.
- The entire model is trained using a supervised learning approach with **Mean Squared Error (MSE) loss**, without requiring prior channel statistics or hardware measurements.
- **Performance** is evaluated under **multiple Signal-to-Noise Ratio** (SNR) conditions and compared with conventional estimation techniques to demonstrate robustness, particularly in low-SNR scenarios.

PROPOSED METHODOLOGY



BLOCK DIAGRAM OF PROJECT FLOW

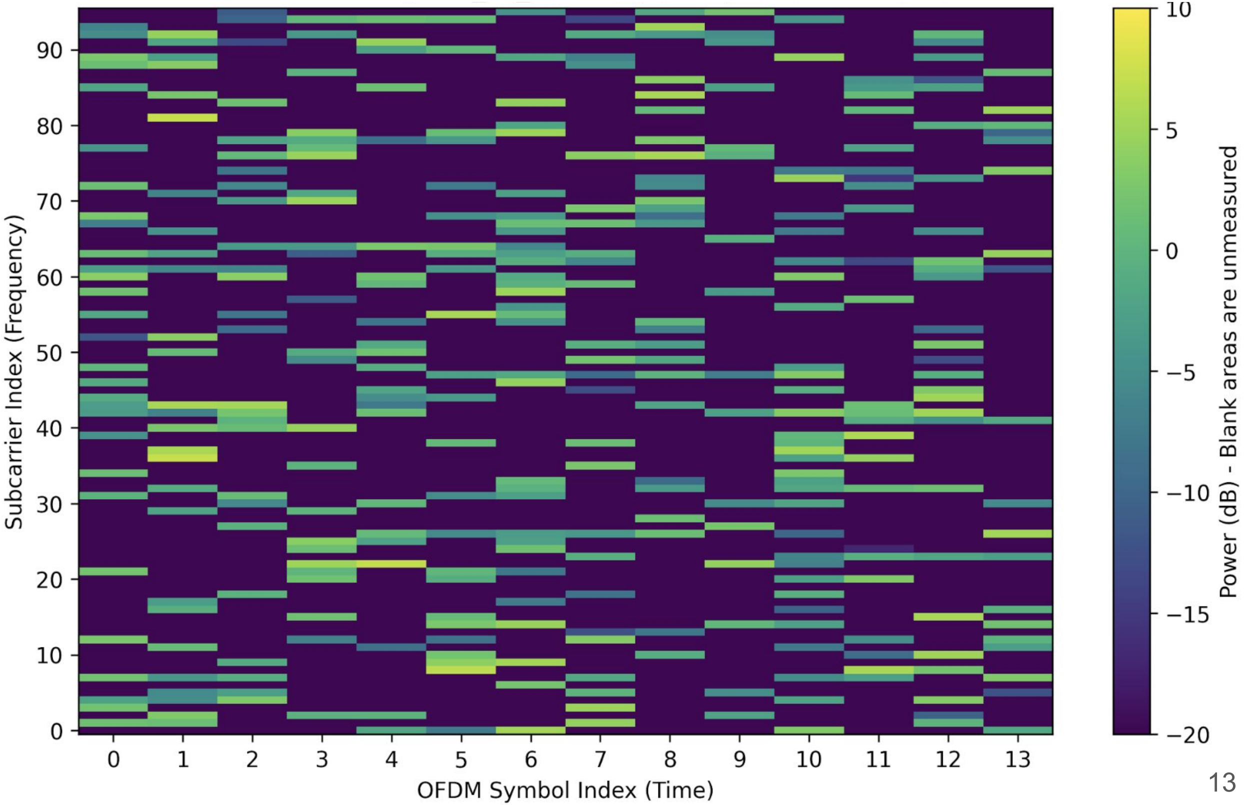


DATASET

Parameters of the dataset:

Parameter	Value
Channel Model	Rayleigh Fading
Multi-Path Profile	Exponential Power Decay
No. of Subcarriers (N_{SC})	96
No. of OFDM Symbols (N_{SYM})	14
Total Resource Elements	$96 \times 14 = \mathbf{1344}$
Pilot Pattern Type	Scattered
Pilot Density	$1/4$ ($\mathbf{25\%}$ of grid)
Modulation	4-QAM / 16-QAM
Performance Metric	NMSE (Normalized MSE)

2D Representation of Data:



PROGRESS:CNN TECHNIQUES

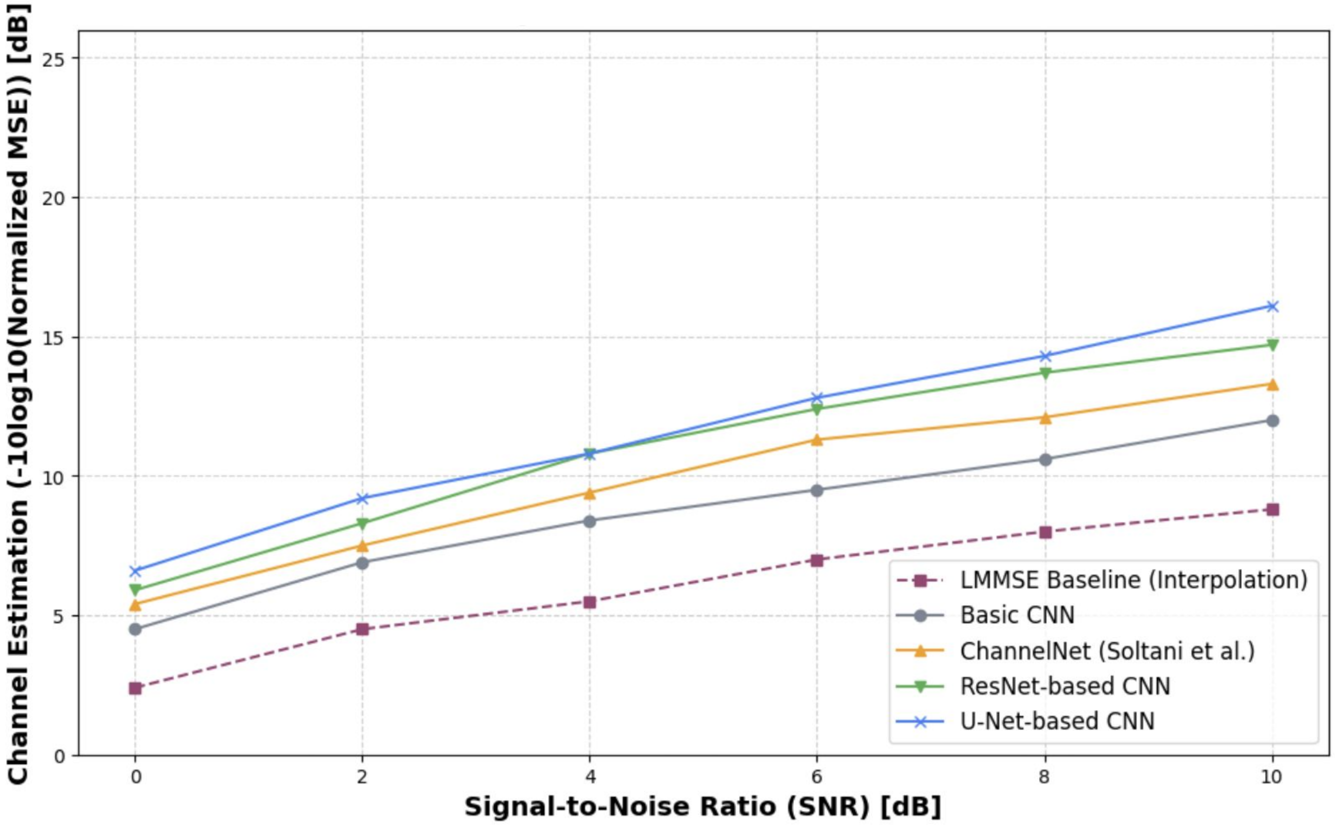
```
==========================================================================================
CNN MODEL COMPARISON TABLE
==========================================================================================
Model                Parameters     Depth/Type                                     Key Connections/Mechanism
------------------------------------------------------------------------------------------
basic_cnn                 185,922 Simple 6-layer CNN (Encoder-Interpolation-Decoder) None (Sequential)
channelnet              1,038,724 SRN (3 blocks) + IRN (5 blocks)                 Residual (within blocks)
resnet                    684,194 8x Residual Blocks (Deep Feature Extraction)    Residual (between Conv layers)
unet                    1,782,978 Encoder-Decoder (2 levels)                      Skip (Concatenation across levels)
attention_cnn             751,234 Encoder-Decoder with SE-style blocks            Channel Attention (Weighting)
==========================================================================================
```

Dataset used:

Parameter	Value
Channel Model	Rayleigh Fading
Multi-Path Profile	Exponential Power Decay
No. of Subcarriers (N_{SC})	96
No. of OFDM Symbols (N_{SYM})	14

Parameter	Value
Total Resource Elements	$96 \times 14 = \mathbf{1344}$
Pilot Pattern Type	Scattered / Sparse
Pilot Density	$1/4$ ($\mathbf{25\%}$ of grid)
Modulation	16-QAM

CNN TECHNIQUES-PERFORMANCE



PRESENT TECHNIQUES

1. Least Square(dB):

$$\hat{H}_{\mathrm{LS}}(k) = \frac{Y(k)}{X_{\mathrm{pilot}}(k)}$$

Estimates the channel by simply **dividing the received pilot signal by the known transmitted pilot symbol**, minimizing the energy of the error in the pilot domain.

2. Linear Minimum Mean Squared Error (LMMSE) (dB):

$$\hat{\mathbf{h}} = \mathbf{C}_{hx} \cdot \mathbf{C}_{xx}^{-1} \cdot \mathbf{x},$$

C*hx***:** Represents the correlation between the true channel and the received signal.
C*xx***:** Represents the correlation of the received signal with itself.

It is a **statistical linear interpolation** method that uses pre-calculated **covariance matrices** to weight received pilot signals and minimize the mean squared error across the channel grid.

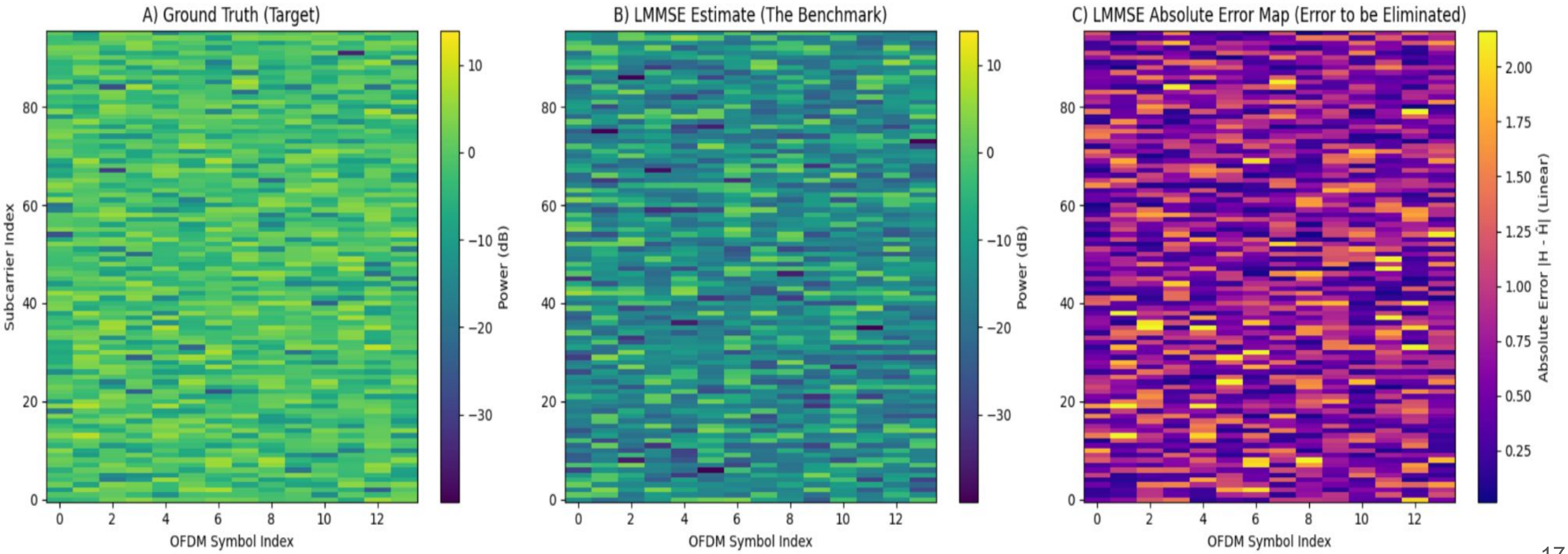
$$\mathrm{NMSE} = 10 \cdot \log_{10}\left(\frac{\mathbb{E}[|H_{\mathrm{True}} - \hat{H}|^2]}{\mathbb{E}[|H_{\mathrm{True}}|^2]}\right)$$

Output:

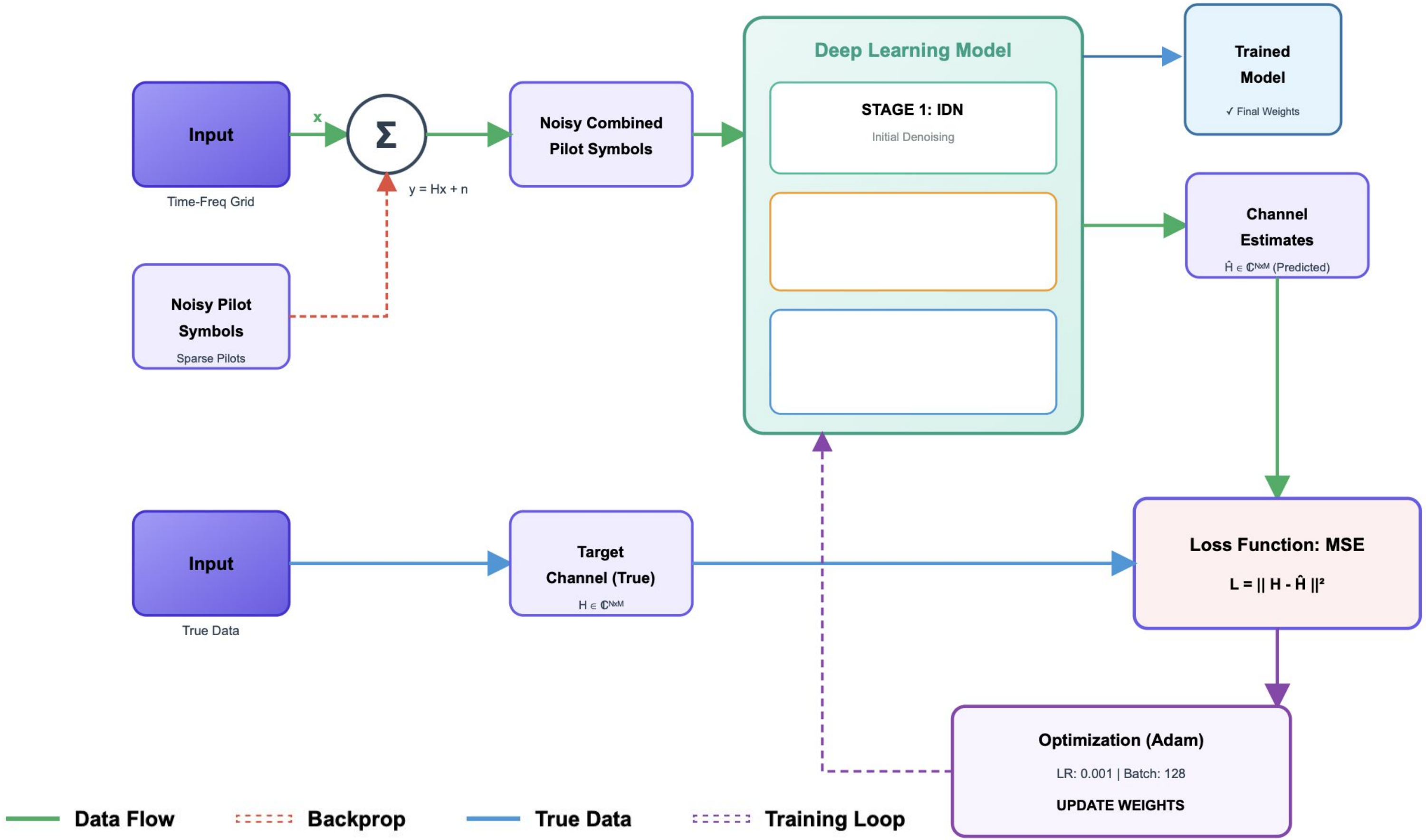
```
--- BENCHMARK RESULTS (SNR = 10 dB) ---
1. LS (Noisy, Sparse Input) NMSE: -1.20 dB
2. LMMSE (Optimal Traditional Method) NMSE: -1.37 dB
---------------------------------------------
```

LMMSE TECHNIQUE

Channel Estimation LMMSE: -1.37dB



FLOWCHART OF INITIAL DENOISING NETWORK



THEORY OF IDN

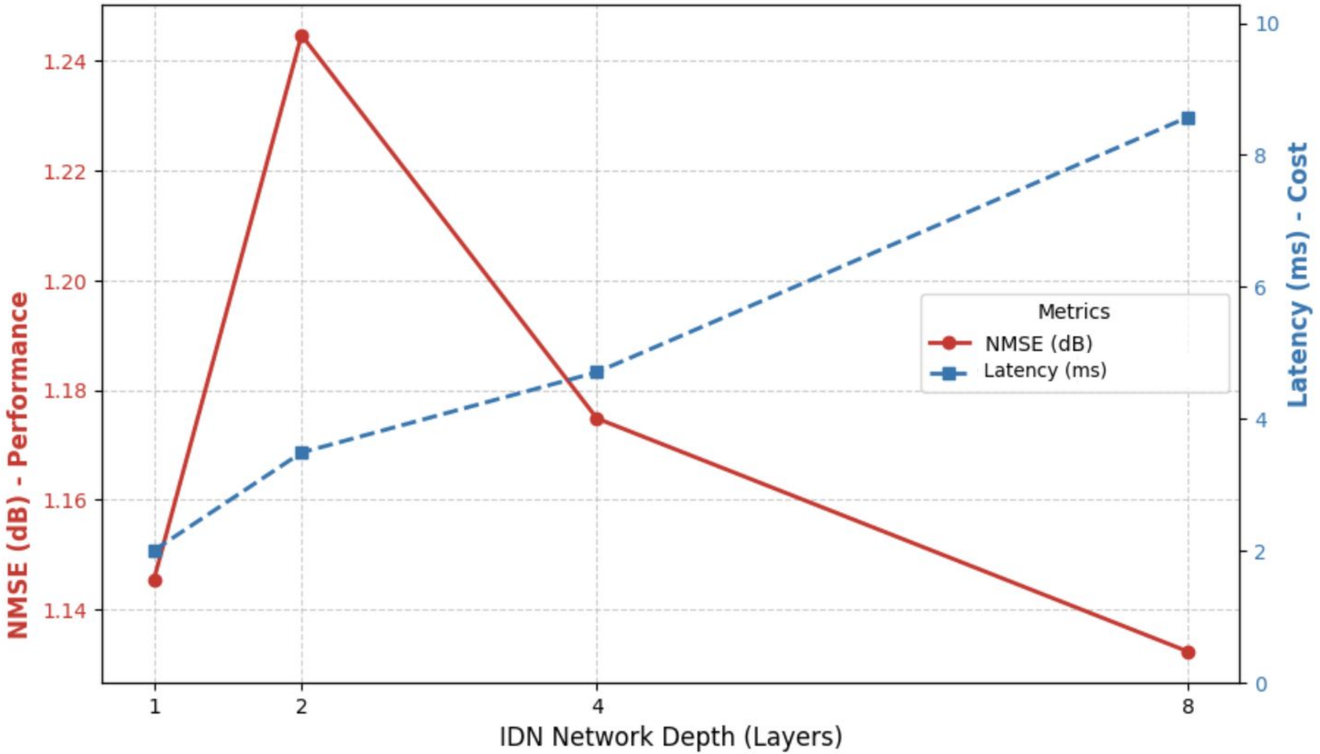
IDN Output Formula : $\hat{\mathbf{H}}_{\text{denoised}} = \mathbf{H}_{LS} - f_{\text{IDN}}(\mathbf{H}_{LS})$

- ➔ $\mathbf{H}_{denoised}$: The **Denoised Channel State Information (CSI)**, which is the final output of the network.
- ➔ $\mathbf{H}_{LS}$: The **Least Squares (LS) Channel Estimate**, which serves as the noisy input to the IDN.
- ➔ $f_{\text{IDN}}(\mathbf{H}_{LS})$: The **Learned Noise Residual** predicted by the 4-layer dense network to be subtracted from the noisy input

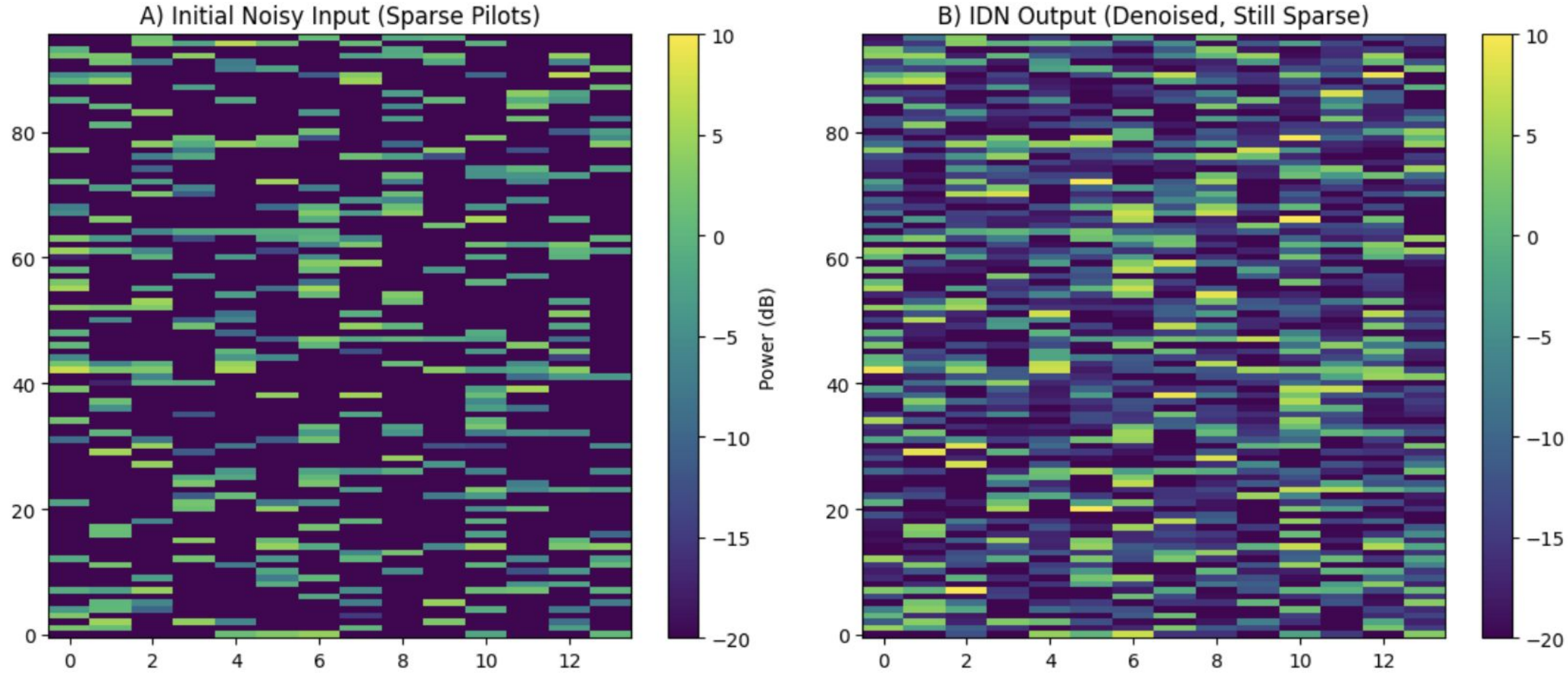
Parameters of the IDN:

Parameter	**Value**
Layers	4 Dense Layers
Activation	LeakyReLU
Normalization	Batch-Normalisation
Output	$\hat{\mathbf{H}}_{\text{denoised}}$

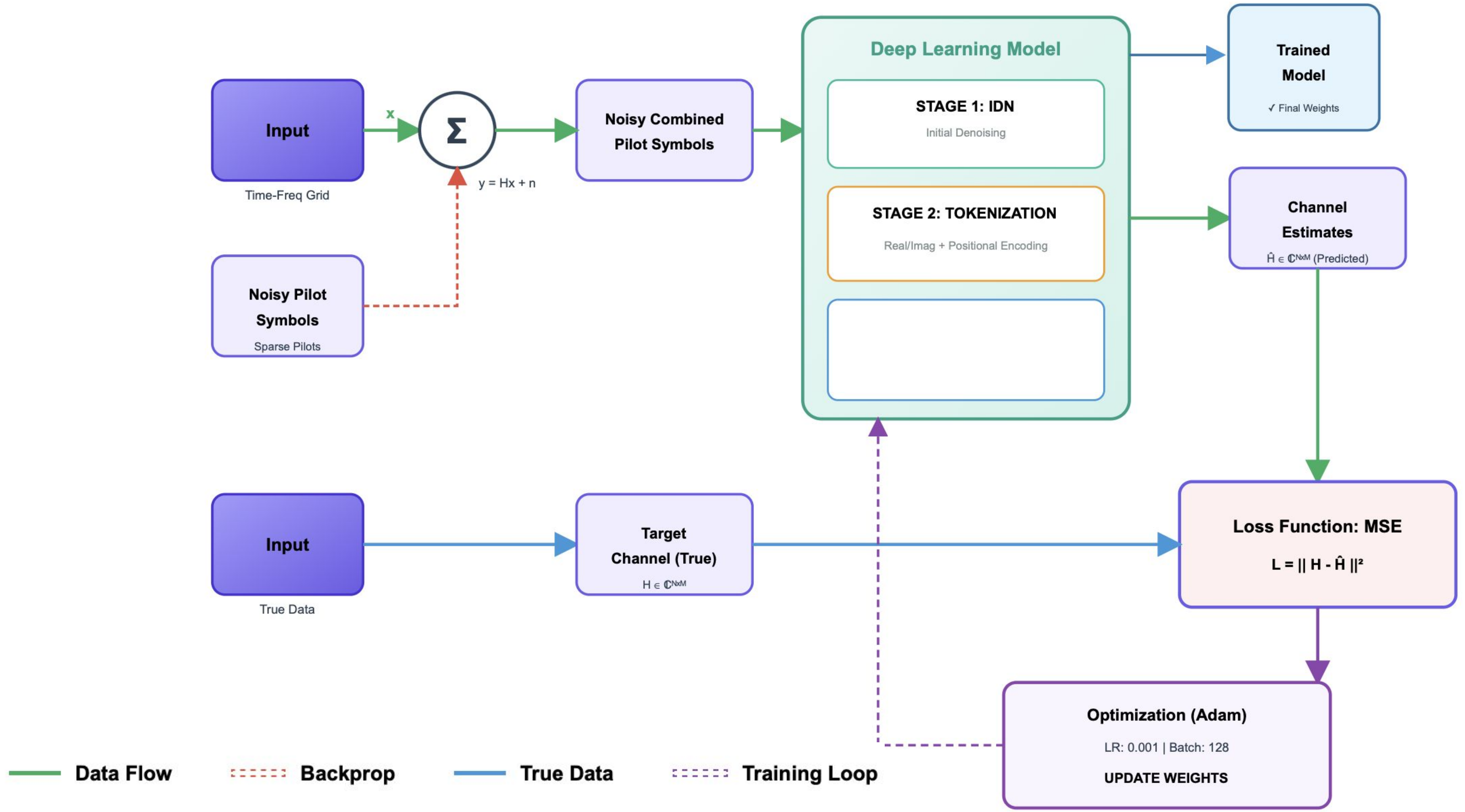
DETERMINATION OF LAYERS IN IDN



INPUT & OUTPUT LAYERS IN IDN



FLOWCHART OF TOKENIZATION



THEORY OF TOKENIZATION

1. Objective

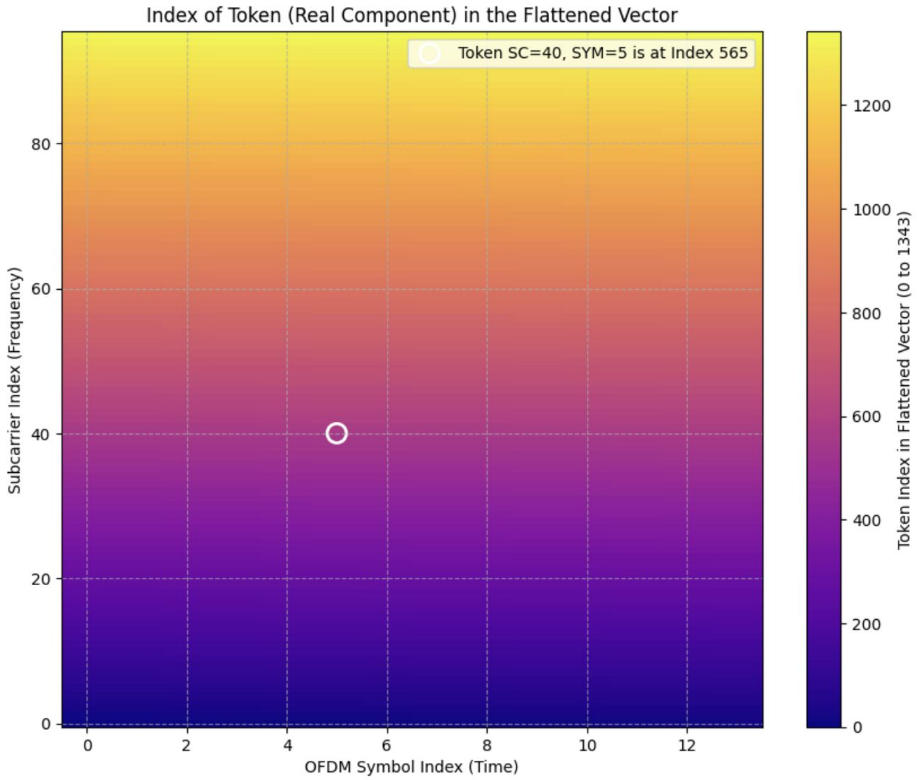
- **Problem:** The Transformer requires **1D sequential data** (like a sentence) but the Channel State Information (**H**) is a **2D grid** (Frequency × Time).
- **Goal:** Convert the 2D channel grid into a structured 1D sequence **without losing the crucial spatial coordinates**.

2. Process

Step	Concept	Description
A. Serialization	Flattening the Grid	The clean pilot data (**H**^denoised) is flattened into a single, long vector (length NSC×NSYM x2).
B. Token Creation	Real/Imag Separation	The complex values are separated into their **Real** and **Imaginary** components, forming the core features of the sequence.
C. Positional Encoding	Injecting Location	Append the original Subcarrier and Symbol index to each element. This tells the Transformer the element's exact location on the grid.

$$\text{Token}_k = [\text{Re}(\hat{H}_k), \text{Im}(\hat{H}_k), \text{Pos}_{\text{freq}}, \text{Pos}_{\text{time}}]$$

2D VISUALIZATION OF TOKENIZATION

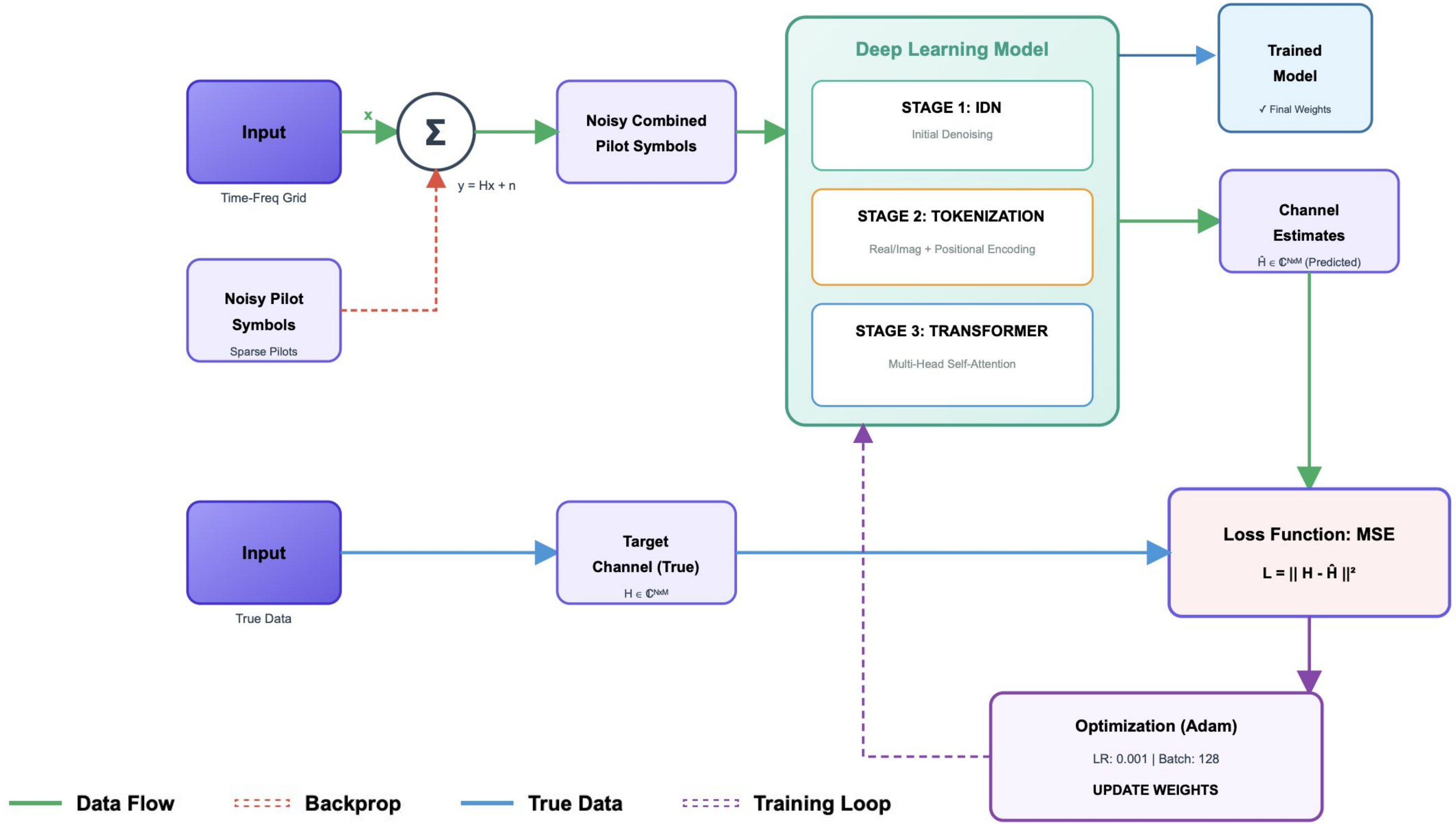


$$\text{Token}_k = [\text{Re}(\hat{H}_k), \text{Im}(\hat{H}_k), \text{Pos}_{\text{freq}}, \text{Pos}_{\text{time}}]$$

The Color Graph

1. **Low Index (Dark Purple/Blue):** These are the elements that come **first** in the sequential vector (e.g., the bottom-left corner).
2. **High Index (Yellow/White):** These are the elements that come **last** in the sequential vector (e.g., the top-right corner).

FLOWCHART OF DEEP LEARNING MODEL



DL MODEL PHASE 1 PARAMETERS

Module	Component	Formula	Parameter Count
IDN	Linear 1	(2688 x 1024) + **512**	2,753,536
	BatchNorm 1	2 x1024	2,048
	Linear 2	(1024 x 512) + **256**	524,800
	BatchNorm 2	2 x 512	1,024
	Linear 3	(512 x 256) + **128**	131,328
	Linear 4	(256 x 2688) + 2688	690,816
Transformer	Linear Embed	(2688 x 512) + 512	1,376,768
	BatchNorm	2 x 512	1,024
	4x Encoder Layers	4 x [dmodel . (12 dmodel + 13)]	~12,600,000
	Output Layer	(512 x 2688) + 2688	1,379,456
TOTAL			**~19.46 Million**

DEEP LEARNING MODEL TRAINING PHASE 1

```
# --- HYPERPARAMETERS ---
BATCH_SIZE = 128
LEARNING_RATE = 0.001
N_EPOCHS = 50
```

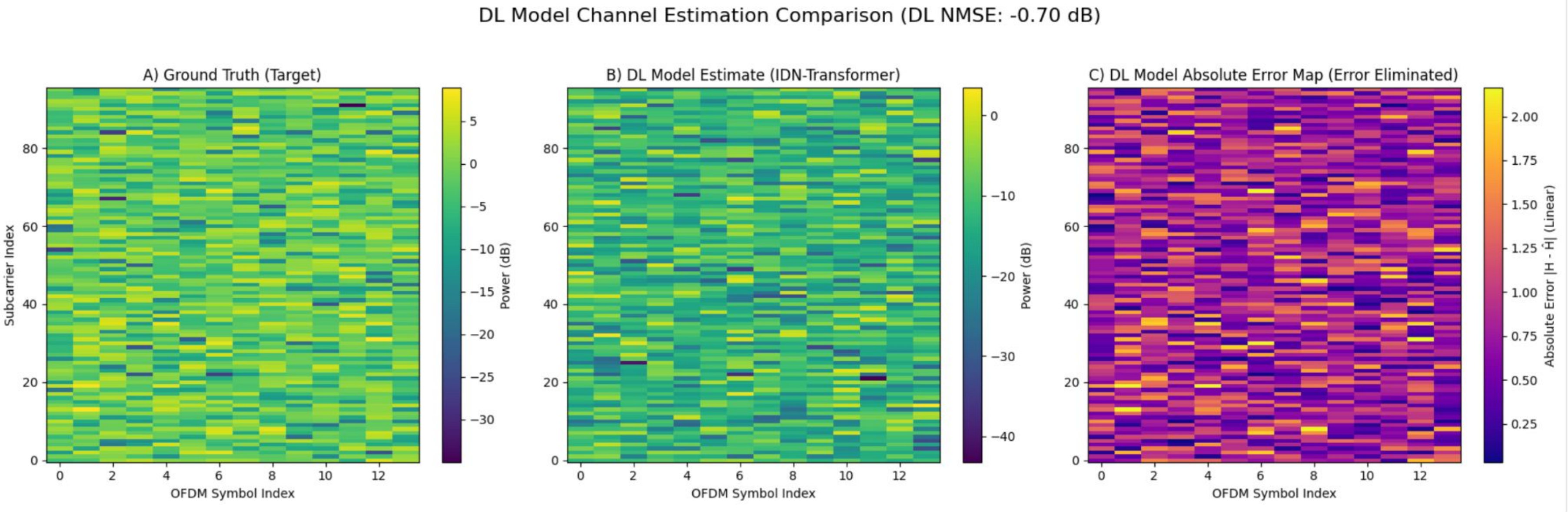
```
--- STARTING TRAINING ---
Epoch [1/50], Loss: 0.527514
Epoch [2/50], Loss: 0.503611
Epoch [3/50], Loss: 0.495742
Epoch [4/50], Loss: 0.485545
```

```
Epoch [43/50], Loss: 0.431157
Epoch [44/50], Loss: 0.430972
Epoch [45/50], Loss: 0.430858
Epoch [46/50], Loss: 0.430710
Epoch [47/50], Loss: 0.430558
Epoch [48/50], Loss: 0.430426
Epoch [49/50], Loss: 0.430358
Epoch [50/50], Loss: 0.430177
```

```
--- FINAL PERFORMANCE RESULTS ---
1. LS (Initial) NMSE:                 -1.20 dB
2. LMMSE (Benchmark) NMSE:            -1.37 dB
3. DL Model (IDN-Transformer) NMSE: -0.70 dB
```

DEEP LEARNING MODEL TRAINING PHASE 1

INPUT AND OUTPUT VISUALIZATION



NMSE (Channel Estimation) = −0.70 dB @ Epoch 50

COMPARISON OF DEEP LEARNING MODEL

Inference:

- **Inference Efficiency:** The **0.039 GFLOPs** per inference indicates a lightweight execution phase, essential for deployment on edge devices.
- **Scalability:** The linear increase in **Total Training GFLOPs** from Phase 1 to Phase 2 shows predictable resource consumption during the development cycle.
- **Real-Time Capability:** The consistent **5ms latency** across both phases confirms the model's stability for time-sensitive resource grid processing (96×14).

1. GFLOPs (Inference Complexity): It tells how much work the computer does to process **one single input** (one channel matrix).

2. Total Training GFLOPs (Computational Energy): It tells the total amount of "math" performed during the entire training process to reach the final weights.

Phase	1	2
Metric	**50 Epochs**	**300 Epochs**
Total Parameters	1,94,61,472	1,94,61,472
Model Size (MB)	74.2 MB	74.2 MB
GFLOPs (Per Inference)	0.039 GFLOPs	0.039 GFLOPs
Total Training GFLOPs*	58,500 GFLOPs	351,000 GFLOPs
Training Time	8 Minutes	40 Minutes
Latency(approx)	5ms	5ms

DEEP LEARNING MODEL TRAINING PHASE 2

```
# --- HYPERPARAMETERS (INCREASED EPOCHS) ---
BATCH_SIZE = 128
LEARNING_RATE = 0.001
N_EPOCHS = 300
```

```
--- STARTING ENHANCED TRAINING (300 Epochs) ---
Epoch [1/300], Loss: 0.528005
Epoch [2/300], Loss: 0.508915
Epoch [3/300], Loss: 0.503364
Epoch [4/300], Loss: 0.493641
```

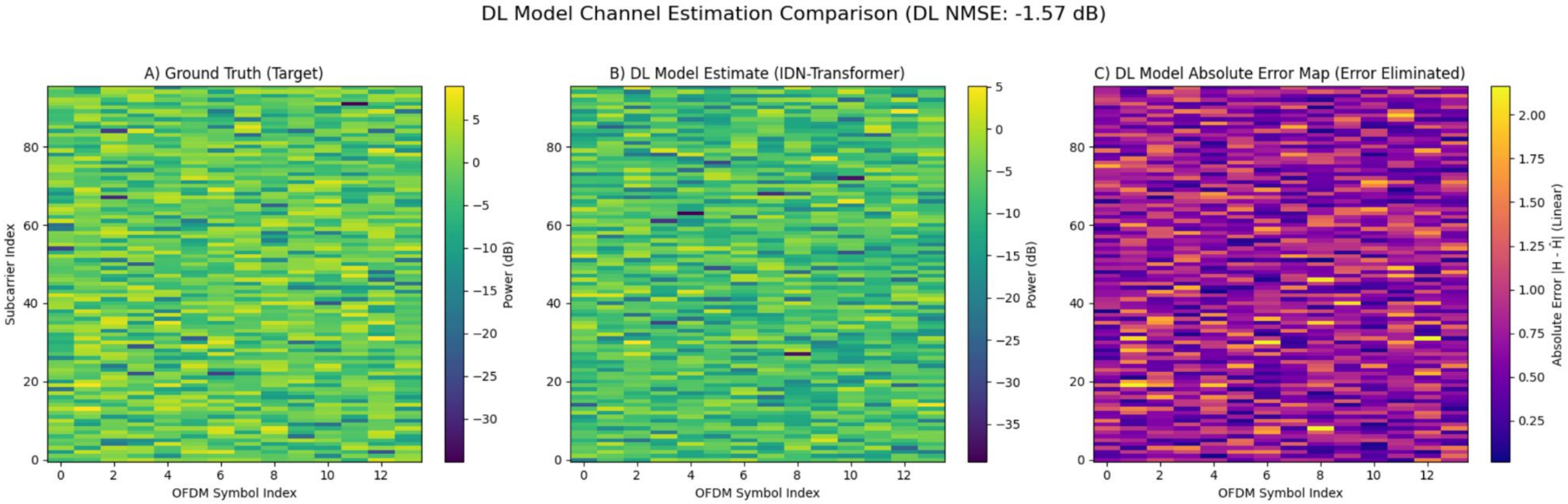
```
Epoch [296/300], Loss: 0.358793
Epoch [297/300], Loss: 0.358903
Epoch [298/300], Loss: 0.358767
Epoch [299/300], Loss: 0.358707
Epoch [300/300], Loss: 0.358711
```

```
--- FINAL PERFORMANCE RESULTS ---
1. LS (Initial) NMSE:                 -1.20 dB
2. LMMSE (Benchmark) NMSE:            -1.37 dB
3. DL Model (IDN-Transformer) NMSE: -1.57 dB
```

DL MODEL PHASE 2 PARAMETERS

Module	Component	Formula	Parameter Count
IDN	Linear 1	(2688 x 1024) + **1024**	2,753,536
	BatchNorm 1	2 x1024	2,048
	Linear 2	(1024 x 512) + **512**	524,800
	BatchNorm 2	2 x 512	1,024
	Linear 3	(512 x 256) + 256	131,328
	Linear 4	(256 x 2688) + 2688	690,816
Transformer	Linear Embed	(2688 x 512) + 512	1,376,768
	BatchNorm	2 x 512	1,024
	4x Encoder Layers	4 x [dmodel . (12 dmodel + 13)]	~12,600,000
	Output Layer	(512 x 2688) + 2688	1,379,456
TOTAL			**~19.46 Million**

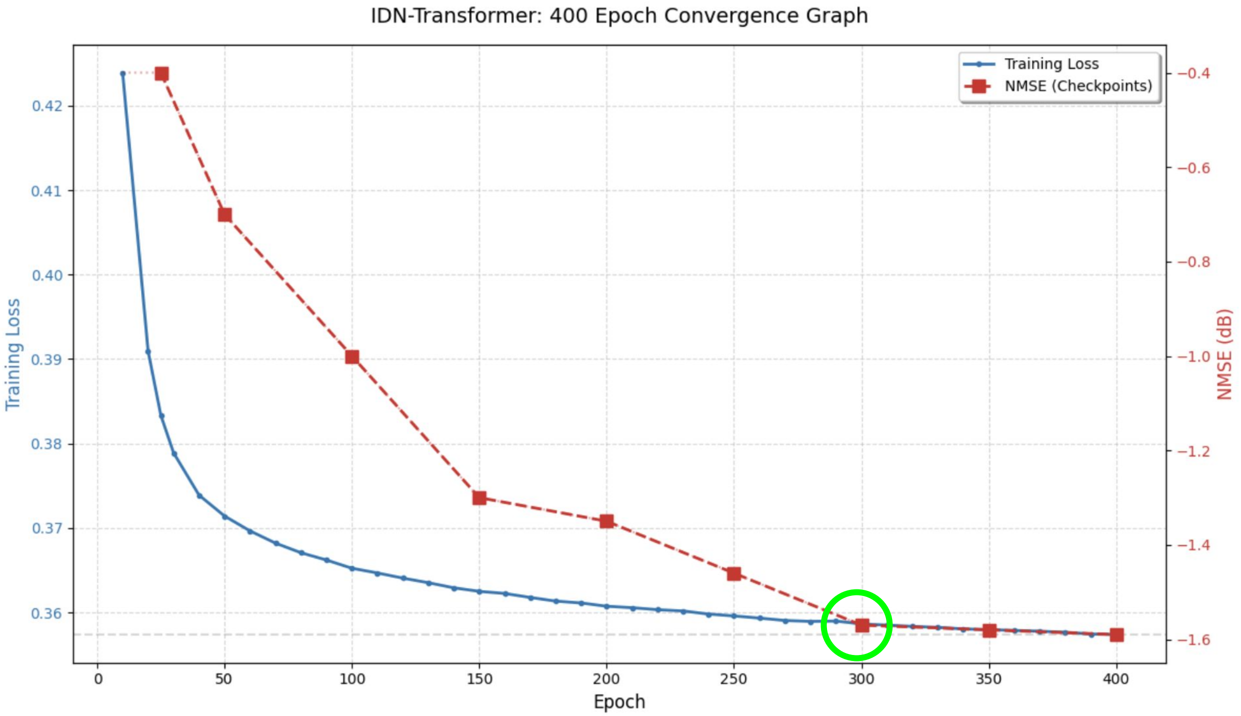
DEEP LEARNING MODEL TRAINING PHASE 2



NMSE (Channel Estimation) = −1.57 dB @ Epoch 300

PROGRESS

JUSTIFICATION DL MODEL TRAINING PHASE 2




```
--- FINAL PERFORMANCE RESULTS ---
1. LS (Initial) NMSE:                 -1.20 dB
2. LMMSE (Benchmark) NMSE:            -1.37 dB
3. DL Model (IDN-Transformer) NMSE: -1.57 dB
```

RESULT

- **Deep Learning Superiority:** The IDN-Transformer consistently outperforms both the LS and LMMSE mathematical baselines under 10 dB SNR conditions.
- **Effective Noise Mitigation:** The Initial Denoising Network (IDN) stage provides a cleaner input, which is particularly critical for sensitive high-order modulations like **16-QAM**.
- **Global Correlation Capture:** By utilizing a **self-attention mechanism**, the model learns complex, non-local time-frequency correlations across the entire **96 × 14 resource grid**, overcoming the local limitations of standard CNNs.

Estimation Method	Model Category	NMSE (dB)	Core Strength
Least Squares (LS)	Mathematical Baseline	-1.20	Standard baseline with high noise sensitivity.
LMMSE	Traditional Optimal	-1.37	Traditional optimal linear estimator.
IDN-Transformer	Proposed Model	**-1.57**	Superior noise suppression & global correlation.

WORKS REMAINING

1.DL model training with [IDN]->[Tokenization]->[Transformer] -For 10000 samples

2.DL model Optimization and switching 16QAM

3.Comparison with Deep Learning model and Linear Minimum Mean Squared Error

SOFTWARE DESIGN

Platform:

Python – Core programming language for OFDM simulation and deep learning implementation

Google Colab – Cloud-based platform for model training and evaluation with GPU support

Libraries used:

PyTorch – Deep learning framework for implementing the denoising network and Transformer model

NumPy & SciPy – Numerical computation and signal processing operations

Matplotlib – Visualization of performance metrics such as NMSE

IMPLEMENTATION

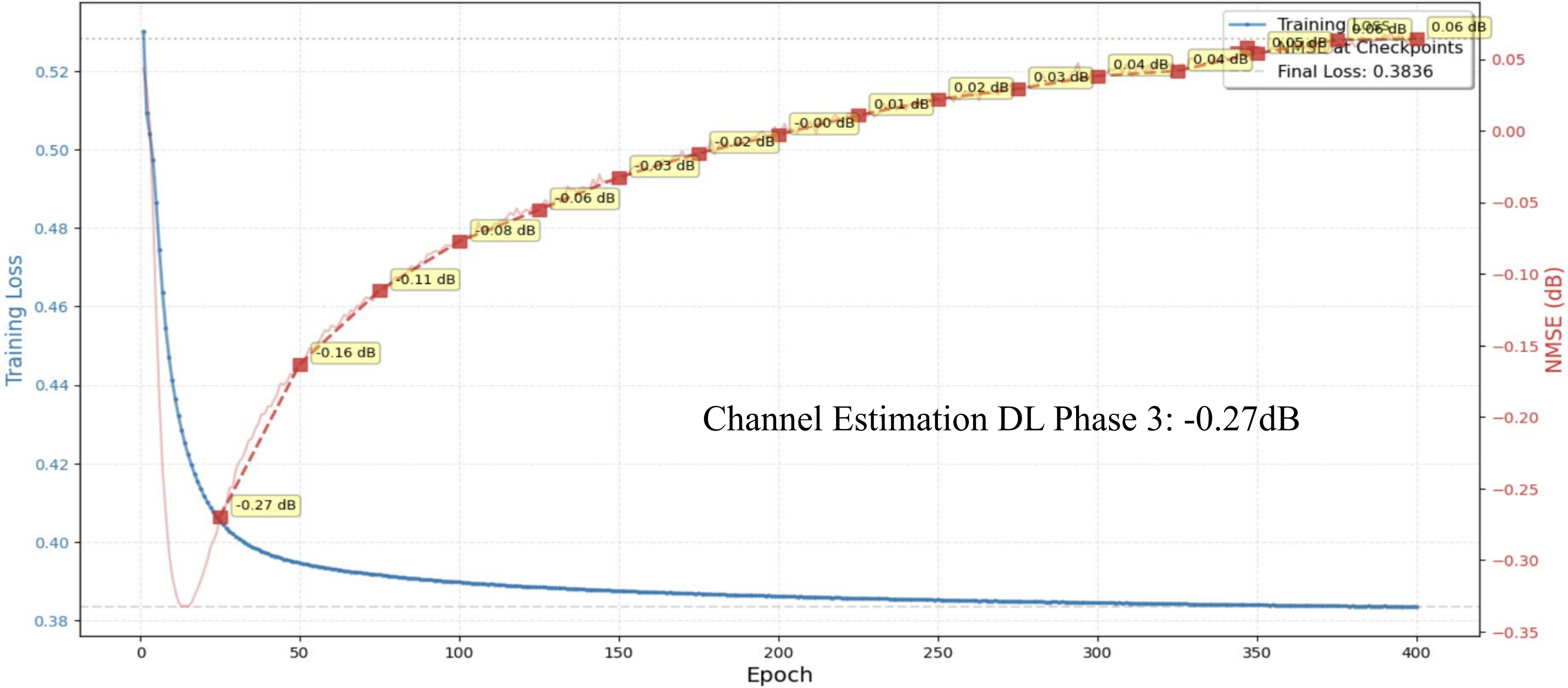
DL MODEL PHASE 3 PARAMETERS REDUCED BY 25%

Module	Component	Formula	Parameter Count
IDN	Linear 1	(2688 × **768**) + **768**	2,065,152
	BatchNorm 1	2 × **768**	1,536
	Linear 2	(768 × **384**) + **384**	295,296
	BatchNorm 2	2 × **384**	768
	Linear 3	(384 × **192**) + **192**	73,920
	Linear 4	(**192** × 2688) + 2688	518,784
Transformer	Linear Embed	(2688 × **384**) + **384**	1,032,576
	BatchNorm	2 × **384**	768
	4x Encoder Layers	4 × [**384** × (12×**384** + 13)]	7,086,336
	Output Layer	(**384** × 2688) + 2688	1,034,880
TOTAL			**9,759,168(49.9% reduction)**

Determination of these Model:

- Preserves the 2:1 ratio between consecutive layers
- Maintains enough capacity for channel estimatian
- Balanced reduction across all layers
- Keeps dimensions multiples of 64 (good for GPU optimization)

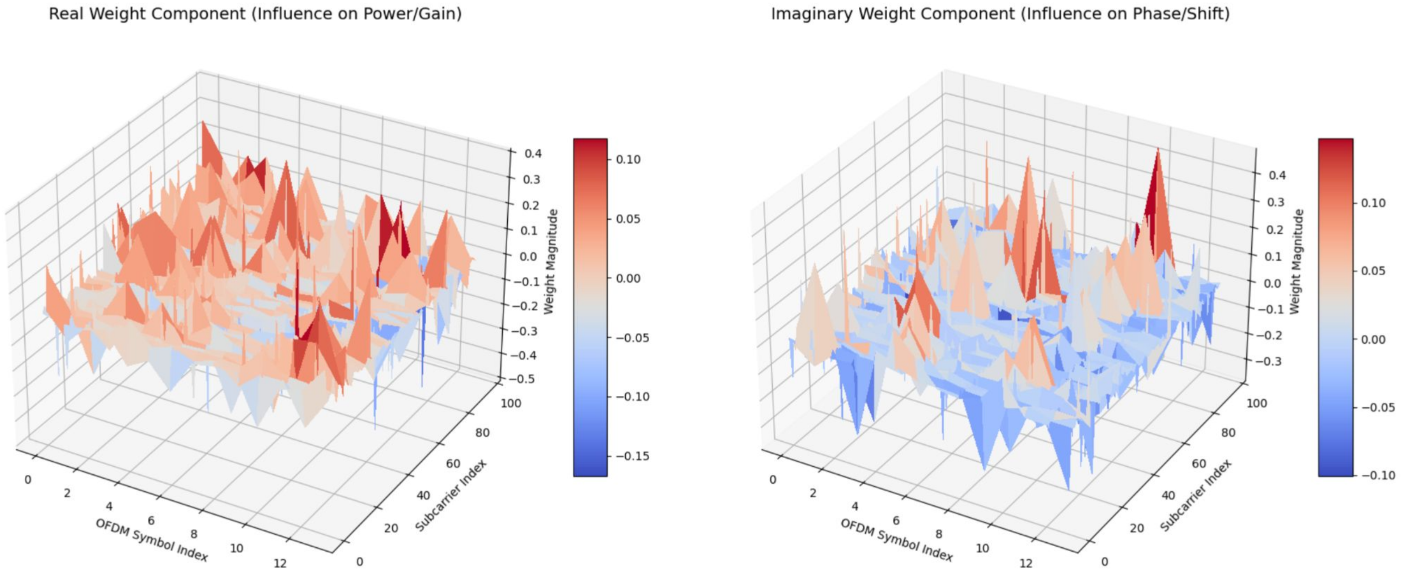
DL MODEL PHASE 3 OUTPUT



DL MODEL PHASE 3 - JUSTIFICATION

1. **Total Parameter Reduction to 50%**
2. **Flattening the Grid**

   Rayleigh fading creates "**peaks**" and "**valleys**" across time and frequency. Adjacent subcarriers and adjacent symbols are highly correlated. By flattening 96 x 14 grid into a single 1D vector and using nn. Linear are completely destroying that 2D geometric relationship. Furthermore, because Transformer's sequence length is forced to 1, **it cannot track how the channel changes over time.**



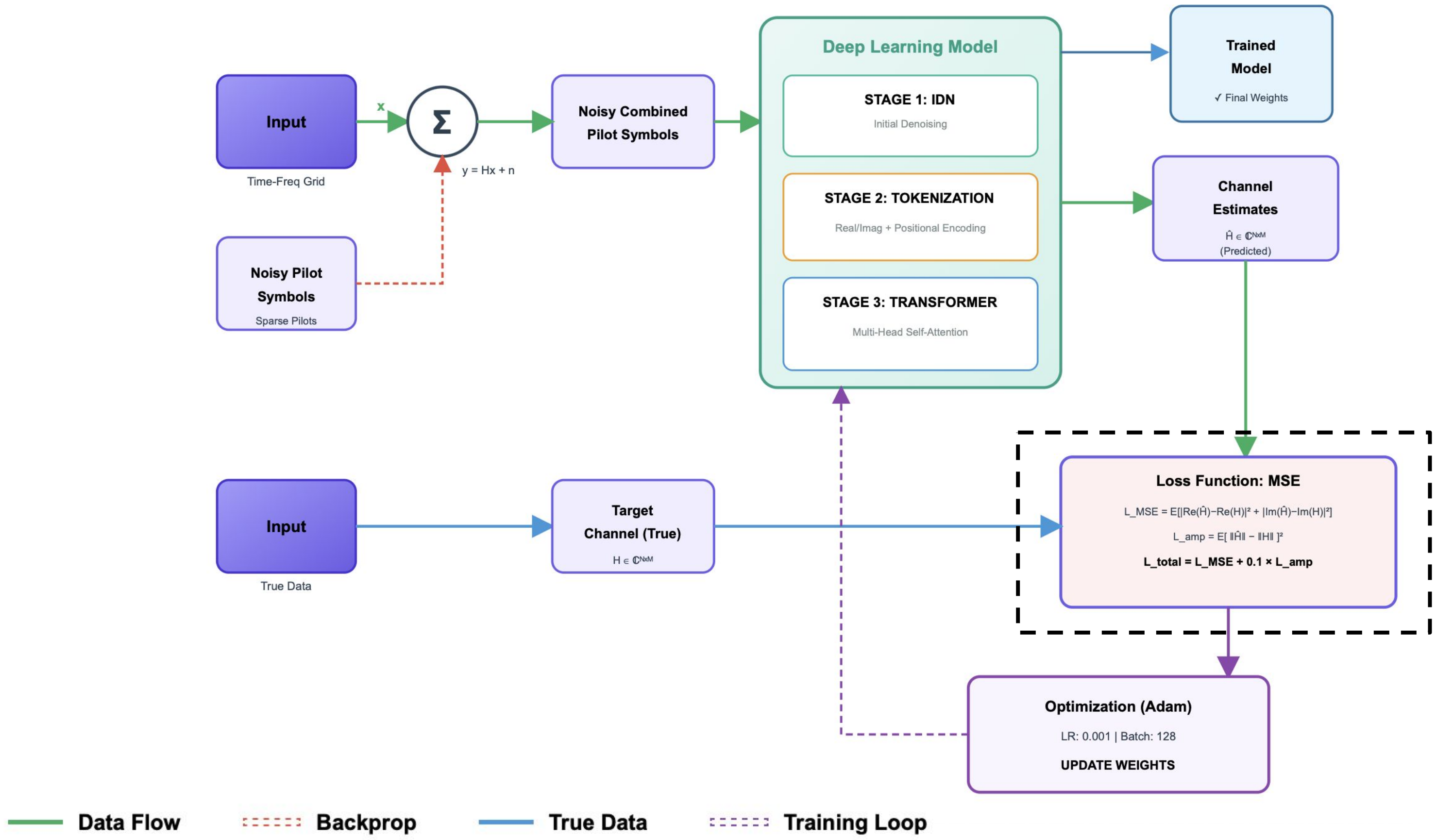
DL MODEL PHASE 3 - VARIETY MODEL

Build and tested model

S No	Total parameters	Total Reduced model	NSME(dB)	Epoch rate	Training time(seconds)
1.	14,838,064	25%	-0.18 dB	40	490 sec
2.	10,697,312	48%	-0.16 dB	35	251 sec
3.	3,268,866	84%	-1.13 dB	90	226 sec
4.	3,268,866 (with dropout-0.3)	84%	-1.17 dB	60	178 sec

The reduction is based on individual parameter reduced to 15%, 25% and 50% from the original parameter used in DL Model Phase 2

DL MODEL PHASE 3 - OPTIMIZATION



DL MODEL PHASE 3 - OPTIMIZATION

Loss Function Upgradation

Current: Standard MSE

$L\_MSE = E[\ |Re(\hat{H}) - Re(H)|^2 + |Im(\hat{H}) - Im(H)|^2\ ]$

- Treats real & imaginary parts independently — no coupling constraint
- A network can minimise MSE while producing wrong complex amplitude |H|
- Received signal power depends on $|H|^2$, so amplitude errors directly hurt SNR
- Phase accuracy is implicit only — no explicit geometric penalty in IQ space

The Fix: Amplitude Regularisation

$L\_amp = E[\ ||\hat{H}|| - ||H||\ ]^2$ $||\cdot|| = \sqrt{(Re^2 + Im^2)}$ (complex magnitude)

$L\_total = L\_MSE + 0.1 \times L\_amp$

- Explicitly penalises wrong signal power — ties directly to receiver SNR
- Weight 0.1 keeps amplitude as a regulariser, not the dominant objective
- Analogous to perceptual losses in image super-resolution
- Preserves physically meaningful channel magnitude profiles

TESTING & VALIDATION

DL MODEL TRAINING PHASE 3 WITH OPTIMIZED

Output:

```
# --- GLOBAL PARAMETERS ---
N_SAMPLES     = 10000
N_SC          = 96
N_SYM         = 14
SNR_dB        = 8
BATCH_SIZE    = 256
LEARNING_RATE = 1e-3
N_EPOCHS      = 400
 Epoch [5], Loss: 0.5758, Val NMSE: 0.32 dB
 Epoch [10], Loss: 0.5328, Val NMSE: -0.21 dB
 Epoch [15], Loss: 0.4953, Val NMSE: -0.74 dB
 Epoch [20], Loss: 0.4613, Val NMSE: -1.31 dB
 Epoch [25], Loss: 0.4284, Val NMSE: -1.86 dB
 Epoch [30], Loss: 0.3998, Val NMSE: -2.40 dB
Epoch [205], Loss: 0.1032, Val NMSE: -5.09 dB
Epoch [210], Loss: 0.1014, Val NMSE: -5.12 dB
Epoch [215], Loss: 0.0996, Val NMSE: -5.08 dB
Epoch [220], Loss: 0.0980, Val NMSE: -5.13 dB
Epoch [225], Loss: 0.0964, Val NMSE: -5.10 dB
Epoch [230], Loss: 0.0950, Val NMSE: -5.07 dB
```

<u>MODEL TRAINING PHASE'S</u>

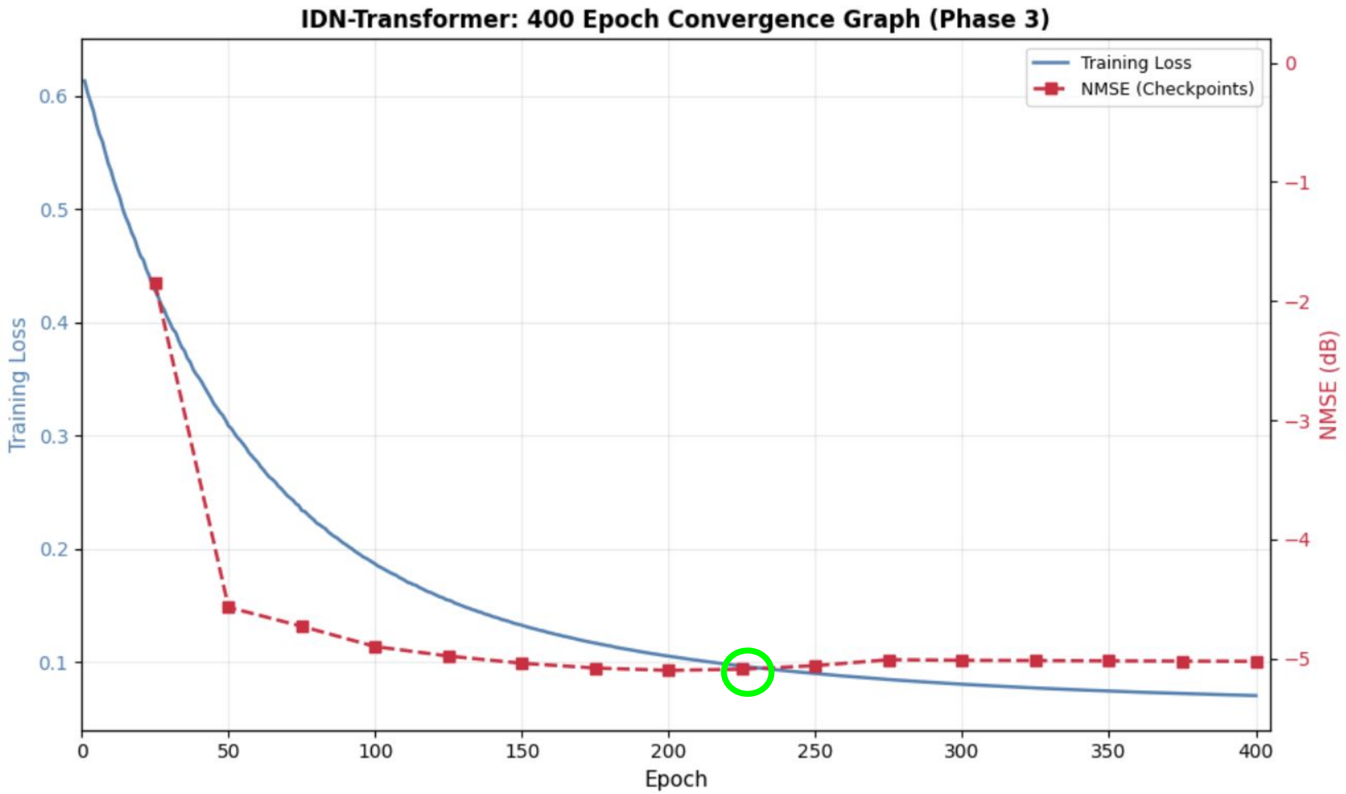
<u>OUTPUT NMSE:</u>

PHASE 1: -0.70dB

PHASE 2: -1.57dB

PHASE 3: -5.10dB

DL MODEL TRAINING PHASE 3 WITH OPTIMIZED



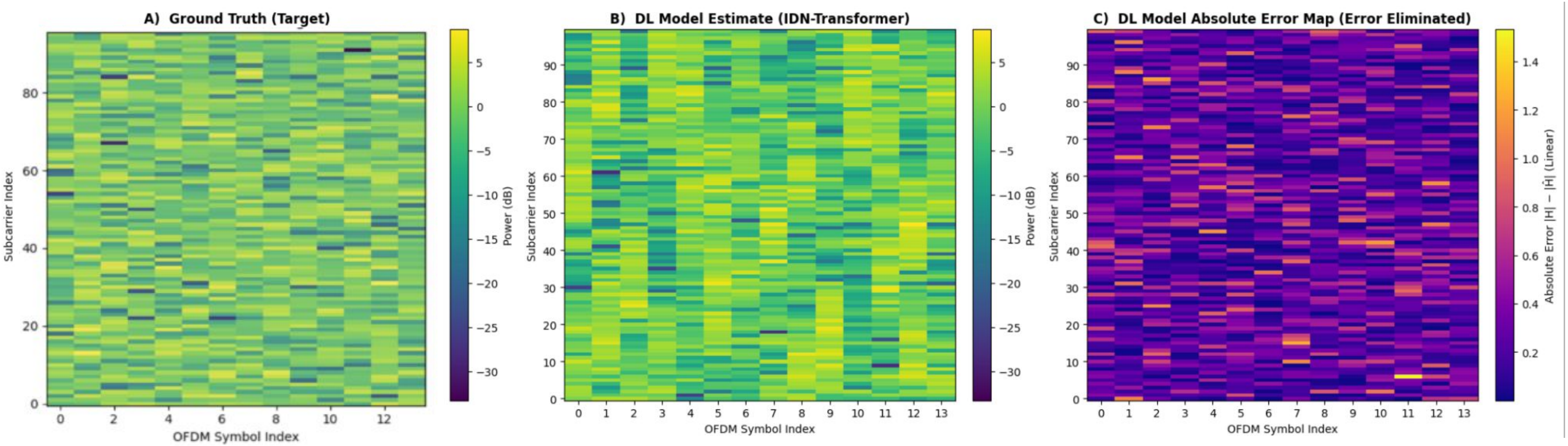
NMSE (Channel Estimation) = −5.10 dB @ Epoch 225

DL MODEL PHASE 3 PARAMETERS

Module	**Component**	**Formula**	**Parameter Count**
IDN	Linear 1	(1024 × 512) + 512	524,800
	BatchNorm 1	2 × 512	1,024
	Linear 2	(512 × 256) + 256	131,328
	BatchNorm 2	2 × 256	512
	Linear 3	(256 × 128) + 128	32,896
	Linear 4	(128 × 1024) + 1024	132,096
Transformer	Linear Embed	(1024 × 256) + 256	262,400
	BatchNorm	2 × 256	512
	2x Encoder Layers	2 × [256 × (12×256 + 13)]	1,579,520
	Output Layer	(256 × 1024) + 1024	263,168
TOTAL			**2,928,256 (3M params)**

DL MODEL TRAINING PHASE 3 WITH OPTIMIZED

Performance Output:



NMSE (Channel Estimation) = −5.10 dB @ Epoch 225

RESULT ANALYSIS & DISCUSSION

Inference:

- **Inference Efficiency:** The 5.86 GFLOPs per inference reflects Phase 3's expanded IDN-Transformer (**3M params**), delivering superior channel estimation at **−5.10 dB NMSE** while remaining suitable for deployment.
- **Scalability:** Total Training GFLOPs scaling to 7,024 GFLOPs in Phase 3 **demonstrates controlled growth**, with each added computation directly justified by measurable NMSE **gains over previous phases.**
- **Real-Time Capability:** The consistent **3 ms latency** in Phase 3 confirms the model's stability for time-sensitive OFDM resource grid processing (96×14), balancing accuracy and deployment efficiency.

1. GFLOPs (Inference Complexity): It tells how much work the computer does to process **one single input** (one channel matrix).

2. Total Training GFLOPs (Computational Energy): It tells the total amount of "math" performed during the entire training process to reach the final weights.

Phase	1	2	3
Metric	**50 Epochs**	**300 Epochs**	**225 Epochs**
Total Parameters	1,94,61,472	1,94,61,472	2,928,256
Model Size (MB)	74.2 MB	74.2 MB	11.18 MB
GFLOPs (Per Inference)	0.039 GFLOPs	0.039 GFLOPs	0.86 GFLOPs
Total Training GFLOPs*	58,500 GFLOPs	351,000 GFLOPs	7,024 GFLOPs
Training Time	8 Minutes	40 Minutes	2 Minutes
Latency(approx)	5ms	5ms	3ms

FURTHER WORKS TO BE COMPLETED

1. DL model Optimization till -10dB with multiple SNR and Justification

2. Comparison with Deep Learning model and Linear Minimum Mean Squared Error

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THANK YOU